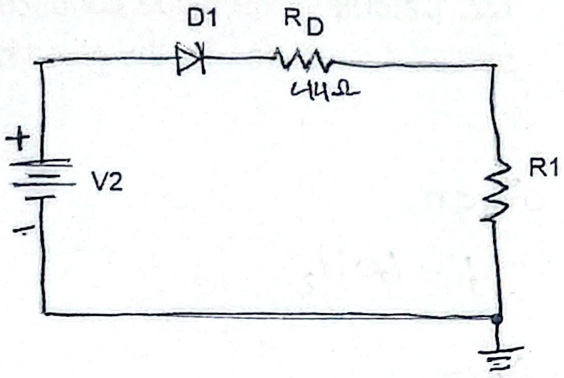
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Q1. Calculate the *value of the load resistor (R1)*, given that the diode's internal resistance (R_D) is 44Ω. The diode current is 1.7969mA when the supply voltage is 8V and the diode current is 1.3138mA when the supply voltage is 6V.



⇒ Given,

$R_D = 44\Omega$

$I_{D_1} = 1.7969\,mA$

$I_{D_2} = 1.3138\,mA$

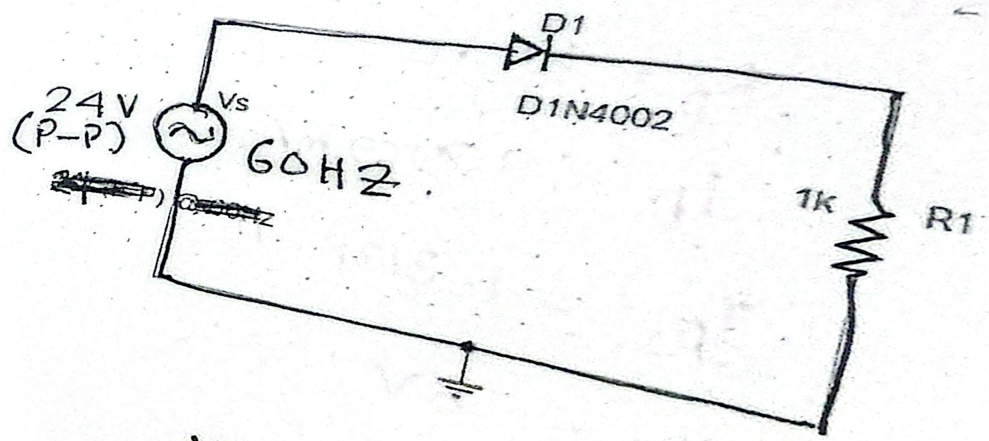
$V_{D_1} = 8V$

$V_1 = 6V$

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Q2. Calculate *the diode conduction percentage, PIV and the average value of the voltage* across the resistor, for the given half wave rectifier circuit.



Given,

$f = 60Hz$

diode conduction percentage:

$$\frac{\frac{T}{2} - 2\Delta t}{T}$$

$$\omega\Delta t = \sin^{-1}\frac{V_m}{V_D}$$

$V_m = 12V$

$V_D = 0.7$

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Q3. **Convert** the given circuit in Q2 into a peak rectifier using a 1mF capacitor, and **calculate** *the ripple voltage (Vr), the average value of the output voltage (Vo), the diode conduction time (Δt), maximum value of the load current, and the average value of the load current.*